

# Vaibhav Patil

Hyderabad | v7patil77@gmail.com | +91 9518941171 | linkedin | github

## Professional Summary

---

Python Developer with 2+ years of experience in backend development using Django, Flask. Skilled in building scalable web applications, RESTful APIs, and microservices with PostgreSQL/MySQL, Redis, Celery. Proficient in AWS, Docker, CI/CD, and Agile practices, with strong debugging and troubleshooting skills.

## Technical skills

---

**Programming Languages:** Python, JavaScript

**Web Framework:** Django, Django REST Framework, Flask, FastAPI, Celery, RabbitMQ, Redis, React JS

**API Development:** RESTful APIs, REST APIs, Micro services, OAuth/JWT Authentication, API Design / Integration

**Databases:** MySQL, SQL, sqlite, postgres, PL/SQL, MongoDB, NoSQL, Redis, RDBMS

**Cloud Technologies:** Docker, Kubernetes (basic), NGINX, Linux, CI/CD, Data processing and Warehousing

**Data Management & Visualization:** PowerBI, Apache Superset, Tableau

**Testing and Debugging:** Postman, Logging, API Testing, Postman, Unit Testing

**Collaboration / Tools:** Webserver, Load Balancer, Git, GitHub Actions, Jenkins, SDLC, Agile, Scrum

**Software Development Practices:** SDLC, Agile, Scrum, Code Reviews, Troubleshooting

## Experience

---

**Project Associate - Backend Developer**, C-DAC – Hyderabad, India

Oct 2024 – Present

- Maintained and enhanced an existing Form-as-a-Service (FaaS) and CForms platform similar to Google Forms, built using **Django** and Django REST Framework (**DRF**).
- Created an interactive analysis dashboard using Streamlit to visualize form insights through donut and bar charts, supporting data-driven event and decision analysis.
- Designed and implemented secure request–response communication using **AES**-based encryption and decryption to protect sensitive form data.
- Strengthened platform security by securing all REST APIs, enforcing best practices for authentication, authorization, and payload validation.
- Developed and integrated role-based access control (**RBAC**) and token-based authentication mechanisms.
- Built backend-to-backend integrations with external services using secure **API keys**, enabling safe and controlled data exchange.
- Integrated asynchronous email notifications using **Celery** and background workers to improve system reliability and responsiveness.
- Performed **API penetration testing** using **Burp Suite**, identified security vulnerabilities, and remediated them using secure code fixes and guard-clause validations.
- Continuously monitored and improved application security and **API security**, addressing potential threats and ensuring compliance with best practices.
- Developed frontend components using React and Vue.js to render dynamic forms.
- Deployed the platform on in-house servers with file storage and SQL database for persistence.
- Engineered RESTful APIs to deliver form services to external applications for registration and feed- back.
- Implemented automated form creation workflows by integrating with external services through API-based triggers, enabling dynamic and seamless form generation across applications.
- Developed **middleware** to validate encrypted payloads, manage file uploads, handle request logging, and monitor API activity.
- Managed and monitored the **Jenkins CI/CD** pipeline, supporting automated builds and deployments for backend services.

**Software Engineer**, Nimble Intelligent Solutions – Nashik, India

March 2024 – June 2024

- Developed high-quality Python **REST APIs** and integrated them with existing business and product workflows.
- Built and deployed **Flask-based** services to support AI application features and platform automation.
- Designed, trained, and deployed machine learning models for **computer vision**, **face recognition**, and pathfinding use cases.
- Implemented interactive **D3.js-based data visualizations**, integrating dynamic charts and graphs into an existing timeline widget within the web application.
- Designed and optimized **SQL** data models and pipelines to support scalable data processing and analytics.
- Created efficient and secure database schemas enabling scalable AI application development.
- Integrated third-party APIs and external services, ensuring compatibility, reliability, and structured data flow across systems.

**AIML Trainee**, PDRL – Nashik, India

July 2023 – Feb 2024

- Designed, automated, and executed **web scraping** workflows to collect large-scale image datasets for model training, using Python-based scraping utilities.
- Scraped and curated satellite imagery from Google Earth and aerial drone imagery, ensuring diversity and relevance for **computer vision** datasets..
- Performed extensive **image annotation and labeling** for drone and satellite data, preparing datasets for computer vision pipelines.
- Organized annotated datasets into **train, validation, and test splits**, ensuring balanced distribution and **model generalization**.
- Implemented **object detection** models using **pre-trained YOLO** architectures, fine-tuning them for aerial and geospatial imagery.
- Applied computer vision techniques to detect objects in satellite and drone imagery, validating model performance against annotated ground truth.

## Education

**Sandip University**, B.Tech - Computer Science and Engineering

May 2019 – May 2023

- GPA: 8.4

## Projects

### FaaS - Form as a Service

[Project Link](#)

web application for managing event-linked forms and organizing responses, with automation features to streamline form creation and reduce manual input for recurring workflows.

Tech Stack: Django DRF, React, Mysql

### - FaaS - Analysis Dashboard

[Project Link](#)

Developed an analytics dashboard for a Form-as-a-Service (FaaS) platform to analyze form submission data in real time. Implemented interactive donut and bar charts to visualize section-wise response metrics and submission counts.

Tech Stack: Django DRF, Streamlit, websocket

### - CForms

[Project Link](#)

A survey and registration tool compatible with Google Forms, used for creating structured forms for events, feedback, and data gathering, supporting a wide range of use cases.

Tech Stack: Django DRF, React, Mysql

## Open Source

---

### - **WebShare**

[Project Link](#)

WebShare is a fast, secure, and lightweight file-sharing solution designed specifically for developers and tech teams. It enables quick file transfers over your local network—no sign-ins, no internet dependency. Your data stays private and within your network.

Tech Stack: Django DRF, React, MongoDB

### - **BotBuilder**

[Project Link](#)

BotBuilder is a full-stack AI agent creation platform that allows businesses to build, train, deploy, and manage custom AI chatbots using their own knowledge base. Unlike generic chatbots, BotBuilder uses Retrieval Augmented Generation (RAG) to ensure that responses are accurate, context-aware, and grounded only in company documents.

Tech Stack: Django DRF, TypeScript, MongoDB